4.2.1.4 I–V characteristics

Some components have a constant resistance. If we double the potential difference, the current doubles. We call these components **ohmic**. Other components do not have a constant resistance – increasing the potential difference might alter the current flow but it does not change proportionately. These are called **non-ohmic**. We can tell which are which by testing them, plotting graphs of the data and looking at the shape of the graph.

Learning outcomes	Maths skills required	Formulae
• Use a range of circuits to safely gather data. • Make measurements using ammeters and voltmeters. • Use a graph to draw conclusions. • Identify variables that need to be kept constant in order to ensure validity of results	• Plot two variables from experimental data. • Interpret graphs that represent a linear relationship.	• potential difference (V) = current (A) × resistance (Ω)

Apparatus list

- 0–12 V variable power supply and connecting leads
- voltmeter or multimeter to measure V
- ammeter or multimeter to measure A
- milliammeter or multimeter to measure mA (you may need this if you have a diode only capable of small currents)
- variable resistor (for example, 10 Ω, 5 A)
- resistor (for example, 100 Ω, 1 W)
- filament lamp (for example, 12 V, 24 W)
- diode
- additional protective resistor (for example, 10 Ω) (optional) – you may need this to ensure your diode does not 'blow'; attach this in series with your diode if advised to by your teacher or science technician
- a switch to turn the circuit on and off (optional)

Safety notes

- Don't turn the power pack up too high! It will damage the components.
- The filament bulb and other components might get hot. Take care not to burn yourself on them.
- Check the equipment before use. If it appears damaged, don't use it.

Common mistakes

- Do not use a.c. in these experiments, use d.c. instead.
- It is **really** hard to set the potential difference to round numbers (e.g. 2 V, 3 V, 4 V) using the variable resistor. If you are struggling, you can take out the variable resistor and use the power pack values for potential difference – even though the potential difference may not be **exactly** what it says on the dial, the amount it increases by is the same each time you turn it up; the intervals are the same.

Diode problems – Activity 3

- If you are getting a current of 0 A or a very low current that doesn't increase at all, it might be that your leads are already switched for negative values. Try swapping them round.
- Don't forget the protective resistor if you have a low current diode! Without the protective resistor, you will blow the diode and then you can't get any results.

Methods

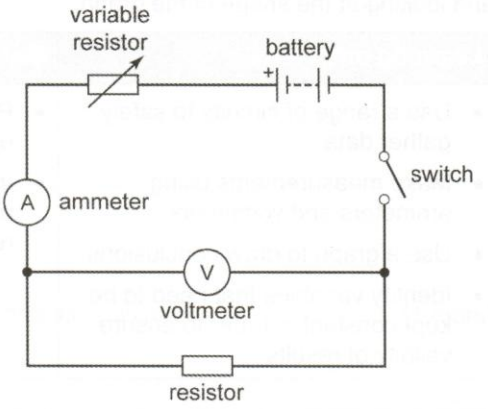
Read these instructions carefully before you start work.

There are three activities to complete.

Activity 1 – Resistor

1. Construct the circuit as shown in **Figure 1**.
2. For different settings of the variable resistor, record the values of current and potential difference. Take five pairs of readings. Turn off the powerpack in between readings to ensure the temperature of the resistor is kept constant.
3. Record the values in **Table 1**.
4. Swap the leads at the power supply (i.e. connect it so that the potential difference is negative).
5. Take readings as before and record them in **Table 1**.

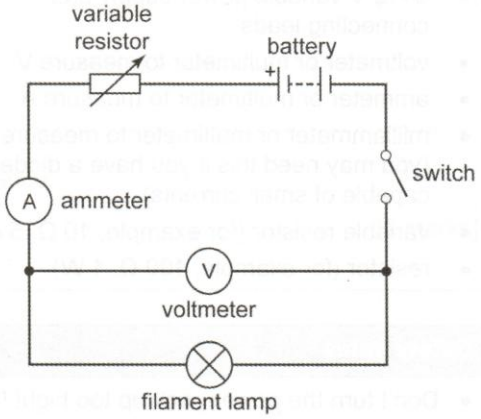
Figure 1



Activity 2 – Filament lamp

1. Construct the circuit as shown in **Figure 2**.
2. For different settings of the variable resistor, record the values of current and potential difference. Take at least five pairs of readings.
3. Record the values in **Table 2**.
4. Swap the leads at the power supply (i.e. connect it so that the potential difference is negative).
5. Take readings as before and record them in **Table 2**.

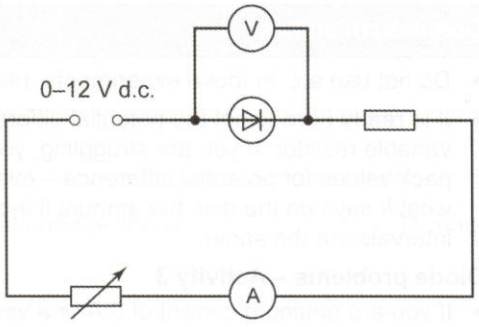
Figure 2



Activity 3 – Diodes

1. Construct the circuit as shown in **Figure 3** using an ammeter if you have a high-current diode, or a milliammeter if you have a low-current diode.
2. This activity may be tricky. Take any readings you can get, even if they are really close together. The range of possible results for this component can be tiny! After you have a few readings, try altering the setting on the variable resistor. Take four further pairs of readings, if possible.
3. Swap the leads at the power supply (i.e. connect it so that the potential difference is negative).
4. Take at least five pairs of readings as before.
5. Record all the values in **Table 3**.

Figure 3



Plot graphs of current against potential difference for each component. Draw lines of best fit. Make sure you label your graphs clearly for each component.

Record your results

Table 1

	Resistor at constant temperature	
	Potential difference (V)	Current (A)
Positive values		
	0	0
Negative values		

Table 2

	Filament bulb	
	Potential difference (V)	Current (A)
	0	0

Table 3

	Diode	
	Potential difference (V)	Current (A)
	0	0